Aman Kumar

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Full-Stack AI Engineer · LangGraph · FastAPI · React · IIT Roorkee '26

PROFESSIONAL SUMMARY

IIT Roorkee graduate with strong full-stack and AI engineering fundamentals: agentic systems, retrieval-augmented generation, and backend API design. Comfortable working across the stack, from Python backends to React frontends, with a focus on reliable, production-grade LLM systems, structured output enforcement, self-checking agents, and gradable evaluation.

EDUCATION

Indian Institute of Technology, Roorkee — Roorkee, Uttarakhand
B.Tech + M.Tech (Integrated Dual Degree), Geophysical Technology — 70% — *2026*

Gaya College (BSEB) — Gaya, Bihar
Intermediate (Class XII) — 77.40% — *2021*

B.L Indo Anglian Public School (CBSE) — Gaya, Bihar
Matriculate (Class X) — 86.80% — *2019*

TECHNICAL SKILLS

AI & Agentic	LangGraph, LangChain, RAG, FAISS, Neo4j, HuggingFace, Groq (Llama 3.1), Anthropic Claude, Prompt Engineering, Structured Output, Vector Embeddings
Backend & APIs	Python, FastAPI, Flask, Node.js, NestJS (Express), REST API Design, JWT Auth, Pydantic, SQLAlchemy, TypeORM
Frontend	React.js, Next.js (App Router), TypeScript, JavaScript, Tailwind CSS
Databases	PostgreSQL, MySQL, MongoDB, SQLite, FAISS, Neo4j
DevOps & Tools	Git, GitHub, Docker, Vercel, Render, Postman, CI/CD concepts

WORK EXPERIENCE

Software Development Intern — Jun 2025 – Jul 2025
CallSure AI

- Built and deployed a LangChain + FAISS retrieval-augmented generation pipeline on Groq Llama, integrated into a live production platform to automate customer support resolution.
- Built and validated a HuggingFace-based classification model against a held-out test set, delivering all sprint work with clean, review-ready code and architecture documentation.

PROJECTS

Agentic Code Evaluation System — Feb – Apr 2025
Python, LangGraph, FAISS, Gradio

- Built a LangGraph ReAct agent with a post-generation validation node that cross-checks findings against explicit rules, eliminating non-deterministic output.
- Designed gradable evaluation infrastructure with sandboxed execution, FAISS-based rubric retrieval, and schema-enforced structured output.

Document Intelligence RAG Platform — Jan – Mar 2025
LightRAG, Neo4j, FAISS, Streamlit

- Built a document intelligence platform combining graph-based and vector-based retrieval to answer questions over large document sets with grounded, cited responses.
- Designed the full pipeline, from document parsing and chunking to embedding, indexing, and retrieval-augmented generation, minimizing irrelevant context passed to the model.

LostFound — Campus Lost & Found Board — Oct – Nov 2024
Python, Flask, SQLite

- Designed and built a full-stack web platform solo, covering item submission, filtering, and a claim/resolve workflow, backed by a secure database schema.
- Deployed for live use across the campus community, resolving issues through direct user feedback and open-sourcing the project with full documentation.

ACHIEVEMENTS

- Codeforces 1550 (Specialist) · CodeChef 2100 (5-Star) · 1500+ algorithmic problems solved.
- Top 1% in JEE Mains and JEE Advanced among 1.2M+ candidates, admitted to IIT Roorkee.